

JAVA SCRIPT

1. JavaScript Basics & Setup

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Community Portal</title>
</head>
<body>

<h1>Community Event Portal</h1>

<script src="main.js"></script>

</body>
</html>
```

main.js

```
console.log("Welcome to the Community Portal");

window.onload = function() {
  alert("Page Fully Loaded");
};
```

2. Syntax, Data Types, and Operators

JavaScript

```
const eventName = "Music Festival";
const eventDate = "10-06-2026";
let seats = 100;
console.log(
  `Event: ${eventName}
  Date: ${eventDate}
  Seats: ${seats}`
);
seats--;
console.log("Available Seats: " + seats);
```

Output

```
Event: Music Festival
Date: 10-06-2026
Seats: 100
Available Seats: 99
```

3. Conditionals, Loops and Error Handling

```
const events = [  
  {name:"Music Fest", seats:10, upcoming:true},  
  {name:"Sports Meet", seats:0, upcoming:true},  
  {name:"Old Event", seats:20, upcoming:false}  
];  
  
events.forEach(event => {  
  
  if(event.upcoming && event.seats > 0){  
    console.log(event.name);  
  }  
});  
  
try{  
  
  let seats = 0;  
  
  if(seats <= 0){  
    throw new Error("No Seats Available");  
  }  
  
}catch(error){  
  
  console.log(error.message);  
}
```

Output

Music Fest
No Seats Available

4. Functions, Scope, Closures, Higher Order Functions

```
function addEvent(name){  
  console.log(name + " Added");  
}  
  
function registerUser(user){  
  console.log(user + " Registered");  
}  
  
function filterEventsByCategory(events, callback){  
  return events.filter(callback);  
}  
  
function registrationTracker(){  
  
  let count = 0;
```

```

    return function(){
        count++;
        return count;
    };
}

const track = registrationTracker();

console.log(track());
console.log(track());

```

Output

```

1
2

```

5. Objects and Prototypes

```

function Event(name,seats){

    this.name = name;
    this.seats = seats;
}

Event.prototype.checkAvailability =
function(){

    return this.seats > 0;
};

let e1 =
new Event("Music Festival",50);

console.log(
e1.checkAvailability()
);

console.log(
Object.entries(e1)
);

```

Output

```

true
[
  ['name','Music Festival'],
  ['seats',50]
]

```

6. Arrays and Methods

```
let events = [];  
  
events.push("Music Show");  
events.push("Workshop on Baking");  
events.push("Music Night");  
  
let musicEvents =  
  events.filter(  
    event => event.includes("Music")  
  );  
  
console.log(musicEvents);  
  
let cards =  
  events.map(  
    event => `Event: ${event}`  
  );  
  
console.log(cards);
```

Output

```
["Music Show","Music Night"]  
  
[  
  "Event: Music Show",  
  "Event: Workshop on Baking",  
  "Event: Music Night"  
]
```

7. DOM Manipulation

HTML

```
<div id="events"></div>  
  
<button onclick="register()">  
  Register  
</button>
```

JavaScript

```
const container =  
  document.querySelector("#events");  
  
const card =  
  document.createElement("div");
```

```
card.textContent =  
"Music Festival";  
  
container.appendChild(card);  
  
function register(){  
  
    card.textContent =  
    "Registered Successfully";  
}
```

8. Event Handling

HTML

```
<select id="category">  
  
<option>Music</option>  
<option>Sports</option>  
  
</select>  
  
<input type="text"  
id="search">  
  
<button id="registerBtn">  
Register  
</button>
```

JavaScript

```
document.getElementById(  
"registerBtn"  
) .onclick = function(){  
  
    alert("Registered");  
};  
  
document.getElementById(  
"category"  
) .onchange = function(){  
  
    console.log(this.value);  
};  
  
document.getElementById(  
"search"  
) .onkeydown = function(){
```

```
    console.log("Searching...");  
  };  
}
```

9. Async JS, Promises, Async/Await

Promise Version

```
fetch(  
  "https://jsonplaceholder.typicode.com/posts"  
)  
  
  .then(response =>  
    response.json())  
  
  .then(data =>  
    console.log(data))  
  
  .catch(error =>  
    console.log(error));
```

Async/Await Version

```
async function loadEvents(){  
  
  try{  
  
    console.log("Loading...");  
  
    const response =  
      await fetch(  
        "https://jsonplaceholder.typicode.com/posts"  
      );  
  
    const data =  
      await response.json();  
  
    console.log(data);  
  
  }catch(error){  
  
    console.log(error);  
  }  
}  
  
loadEvents();
```

10. Modern JavaScript Features

```
const events = [  
  {
```

```

        name:"Music Fest",
        seats:50
    }
];

function showEvent(
name = "Unknown Event"
){

    console.log(name);
}

const {
    name,
    seats
} = events[0];

console.log(name);
console.log(seats);

const clonedEvents =
[...events];

console.log(clonedEvents);

```

11. Working with Forms

HTML

```

<form id="regForm">

<input
name="username"
placeholder="Name">

<input
name="email"
placeholder="Email">

<select name="event">
<option>Music</option>
<option>Sports</option>
</select>

<button>
Submit
</button>

<p id="error"></p>

```

</form>

JavaScript

```
document.getElementById(
  "regForm"
).addEventListener(
  "submit",

  function(event){

    event.preventDefault();

    let form =
    event.target;

    let name =
    form.elements.username.value;

    let email =
    form.elements.email.value;

    if(name==="" || email===""){

      document.getElementById(
        "error"
      ).innerHTML =
      "All fields required";

      return;
    }

    alert("Registration Success");
  });
```

12. AJAX & Fetch API

```
function register(){

  setTimeout(async()=>{

    try{

      const response =
      await fetch(
        "https://jsonplaceholder.typicode.com/posts",
        {
          method:"POST",
```



```
headers:{
  "Content-Type":
  "application/json"
},

body:JSON.stringify({
  name:"Ram"
})
});

if(response.ok){

  alert(
    "Registration Successful"
  );

} else {

  alert(
    "Registration Failed"
  );
}

} catch(error){

  console.log(error);
}

},2000);

}

register();
```

13. Debugging and Testing

JavaScript

```
function submitForm(){

  console.log(
    "Form Submission Started"
  );

  let name = "Ram";

  console.log(
    "User Name:",
    name
```

```
);  
  
debugger;  
  
fetch("/register")  
  .then(response =>  
    console.log(response));  
  
}
```

14. jQuery and JS Frameworks

HTML

```
<script src=  
"https://code.jquery.com/jquery-3.7.1.min.js">  
</script>
```

```
<button id="registerBtn">  
Register  
</button>
```

```
<div id="card">  
Community Event  
</div>
```

jQuery

```
$(document).ready(function(){  
  
  $("#registerBtn").click(function(){  
  
    alert("Registered");  
  
    $("#card").fadeOut(1000);  
  
    $("#card").fadeIn(1000);  
  
  });  
  
});
```